

A two-replica proof of the Krenn–Gu conjecture

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Abstract

We consider finite loopless multigraphs whose edges have complex weights and whose two half-edges may have different colours. Each perfect matching induces a vertex colouring, and the weight of a vertex colouring is the sum of the weights of the matchings inducing it. The Krenn–Gu conjecture excludes graphs on more than four vertices for which at least three monochromatic colourings have nonzero weight and every other colouring has weight zero. This manuscript presents a proof based on two formal replicas of the matching polynomial. The argument first eliminates aggregate bichromatic edge weights and then establishes a supported-edge hafnian identity. This identity forces each active colour in the aggregate graph to be a perfect matching, reducing the conclusion to a known combinatorial obstruction. All intermediate arguments are included.

Manuscript status. Research manuscript, September 26, 2026. The argument has undergone two internal research-agent audits. External peer review and a complete Lean formalization remain pending. The mathematical source and its review record are archived in [11]. The sources and the limits of formal verification are stated in Section 8.

1 Introduction

The graph formulation of the Krenn–Gu problem assigns a complex amplitude to each vertex colouring by summing over perfect matchings. Krenn, Gu and Soltész formulated the corresponding existence question in [1, Question 1]. We use the general complex weighted model, including parallel edges and edges with distinct endpoint colours. Our terminology follows the graph formulation in Chandran, Gajjala and Illickan [2, Sections 1.1 and 1.4], who state the conjecture as their Conjecture 6.

1.1 Preliminaries and notation

Let $G = (V, E)$ be a finite loopless multigraph. Each edge $e = uv$ has weight $w(e) \in \mathbb{C}$ and two endpoint colours $\chi_u(e), \chi_v(e)$. We write $[d] = \{1, \dots, d\}$ for the target palette and $\mathcal{M}(G)$ for the set of perfect matchings of G . For $M \in \mathcal{M}(G)$, let $e_M(v)$ be its unique edge at v . The matching induces the vertex colouring $v \mapsto \chi_v(e_M(v))$. For a vertex colouring $\kappa : V \rightarrow [d]$, define

$$W(\kappa) = \sum_{\substack{M \in \mathcal{M}(G) \\ \chi_v(e_M(v)) = \kappa(v) \text{ for every } v}} \prod_{e \in M} w(e).$$

An empty sum is zero. We write i^V for the colouring that is constantly i . A colouring that is not constant is called mixed. For a matrix B and a vertex set S , $B[S]$ denotes its principal submatrix on S .

Definition 1.1 (Generalized GHZ condition). The weighted half-edge-coloured graph satisfies the generalized GHZ condition on $[d]$ if $W(i^V) = \tau_i \neq 0$ for every $i \in [d]$, and $W(\kappa) = 0$ for every mixed vertex colouring $\kappa : V \rightarrow [d]$.

The usual normalized condition requires $\tau_i = 1$. The scaling lemma of [2, Lemma 14] relates the two formulations. We work directly with nonzero τ_i , so no normalization step is needed in the proof.

1.2 Related work

Chandran, Gajjala and Illickan prove the conjecture for graphs of vertex connectivity at most two and for graphs of maximum degree at most three. Their reduction also implies that a counterexample with the fewest vertices must be four-connected [2, Theorems 9–11]. These results concern the same weighted model.

A separate result, attributed to Bogdanov in [2, Theorem 7], states that three monochromatic perfect matchings of distinct colours on more than four vertices force a mixed perfect matching. This is an existence statement. In the complex weighted setting, different matchings may cancel, so it does not by itself settle the conjecture. Our structural argument removes this possibility before invoking the combinatorial obstruction.

1.3 Main result and proof overview

Theorem 1.2. *If $|V| > 4$ is even, there is no complex weighted half-edge-coloured multigraph satisfying the generalized GHZ condition on a palette of size $d \geq 3$.*

It suffices to consider three active colours: deleting all other endpoint colours preserves the amplitudes of the retained vertex colourings. We express these amplitudes in a commutative algebra in Section 2. The proof then has three steps.

First, the reflection argument proves Lemma 3.1. Lemmas 4.1 and 4.2 then force every aggregate edge matrix to be diagonal.

Second, for a diagonal colour matrix B , Theorem 5.1 proves

$$\text{haf}(B) = B_{pq} \text{haf}(B[V \setminus \{p, q\}]) \quad \text{for every } B_{pq} \neq 0.$$

Orthogonal covariance and two original root identities reduce this assertion to polynomial differential equations, resolved by Lemma 5.3.

Finally, hafnian expansion forces each aggregate colour graph to be a perfect matching. Cancellation is then impossible, and Lemma 7.1 completes the contradiction. The four-vertex example remains permitted.

2 Algebraic formulation and formal pairings

Theorem 2.1 (Algebraic form of Theorem 1.2). *Let V have even cardinality $n > 4$. At each vertex take three designated coordinates a, b, c . In the commutative algebra in which the product of any two coordinates at the same vertex is zero, let A be any quadratic involving only distinct vertices, with arbitrary complex coefficients. Write $A^{[j]} = A^j/j!$. Then*

$$A^{[n/2]} = \tau_a a^V + \tau_b b^V + \tau_c c^V, \quad \tau_a \tau_b \tau_c \neq 0 \tag{1}$$

is impossible. Here h^S means the product of coordinate h over S .

We explain the relation with the graph formulation. Fix an ordering of V . Let $A_{uv}(i, j)$ be the sum of the weights of edges whose endpoint colours are i at u and j at v . Replace each such parallel class by one edge of its total weight, deleting it if that total is zero. Expanding products shows that this aggregation preserves every $W(\kappa)$. Henceforth all support, colour-degree and uniqueness statements refer to this aggregate weighted graph. In the algebra

$$\mathcal{A}_V = \mathbb{C}[x_{v,i} : v \in V, i \in \{a, b, c\}] / (x_{v,i}x_{v,j} : v \in V, i, j \in \{a, b, c\}),$$

put

$$A = \sum_{u < v} \sum_{i, j \in \{a, b, c\}} A_{uv}(i, j) x_{u,i} x_{v,j}.$$

Every surviving term in its divided top power chooses each vertex exactly once. The division by $(n/2)!$ removes the ordering of the chosen edges. Thus the coefficient of $\prod_v x_{v, \kappa(v)}$ is exactly $W(\kappa)$. This proves that the tensor condition in (1) is equivalent to the ternary instance of Definition 1.1. Parallel edges of identical endpoint colours aggregate by addition. Projecting any larger target palette onto three active colours preserves (1), so the result covers every target dimension at least three. Unequal nonzero pure amplitudes, arbitrary edge density, and arbitrary complex weights are allowed.

We use formal Wick moments, defined as sums over pairings of labeled factors with assigned symmetric covariances. Odd moments are zero. These are polynomial identities; they require neither positivity nor an invertible covariance. The pairing formula is the algebraic version of the classical moment formula of Isserlis [3]. All uses of covariance below follow directly from multilinearity and the definition as a sum over pairings. The proof of Theorem 2.1 is completed in Sections 6 and 7.

3 Whole binary higher responses vanish

The following lemma combines the reflection and common-factor argument of [5], the two-colour mixed-coefficient identity of [6], and the finite-degree pure-response argument of [7]. We give the complete proof, including the terminal response.

Lemma 3.1 (Whole binary higher-response vanishing). *Let Ω have $2m + 1 \geq 3$ vertices, let R be an arbitrary quadratic on Ω , and let D be a vector space of rows L such that*

$$\Phi(L) = LR^{[m]} = \sum_h f_h(L) h^\Omega. \quad (2)$$

Assume all three linear functions f_h are nonzero and at least two are independent. For odd $p \leq 2m + 1$, define

$$H_p(L) = L^p R^{[(2m+1-p)/2]}.$$

For every $p \geq 3$, the tensor $H_p(L)$ has zero coefficient on every word using at most two colours, including the terminal degree $p = 2m + 1$. The same vanishing holds after polarization for products of arbitrary rows from the same D .

Proof. Give formal variables $X_{i,h}$ their R covariances, zero covariances within a vertex, and give an auxiliary g covariance $L_i(h)$ with $X_{i,h}$ and zero covariance with itself. A Wick moment with p

copies of g and one specified coordinate at each vertex is the corresponding coefficient of $H_p(L)$. The $p!$ assignments of auxiliary occurrences are precisely those of the raw power L^p ; the divided power of R counts each remaining matching once.

Take two independent identical replicas $(X, g_1), (Y, g_2)$. Choose distinct local colours α_i, β_i at each vertex and set

$$\Delta_i = X_{i, \alpha_i} Y_{i, \beta_i} - X_{i, \beta_i} Y_{i, \alpha_i}.$$

Over $\mathbb{C}(s, t)$, the orthogonal reflection

$$\frac{1}{s^2 + t^2} \begin{pmatrix} s^2 - t^2 & 2st \\ 2st & t^2 - s^2 \end{pmatrix}$$

fixes $sg_1 + tg_2$ and negates each Δ_i . It preserves the replicated covariance, so the moment of

$$(sg_1 + tg_2)^{p+1} \prod_{i \in \Omega} \Delta_i$$

is zero: there are an odd number of determinants. Extracting $s^p t$ gives

$$\sum_{I \subseteq \Omega} (-1)^{|I|} H_p(L)(w_I) \Phi(L)(w'_I) = 0, \quad (3)$$

where w_I uses β on I and α elsewhere, and w'_I is complementary. Equality over $\mathbb{C}(s, t)$ gives a polynomial identity, including isotropic parameter values by extension. No operation on the original source is being made.

With constant local pair a, b , (3) says

$$f_b[a^\Omega] H_p = f_a[b^\Omega] H_p.$$

Two independent f 's are coprime in $\mathbb{C}[D]$, hence for $p = 2r + 1$ there is a common homogeneous polynomial q_r of degree $2r$ such that $[h^\Omega] H_p = f_h q_r$ for every h ; $q_0 = 1$. For a mixed word w avoiding an active colour h , choose $\alpha_i = w_i, \beta_i = h$. The only constant complementary word in (3) is h^Ω , giving $f_h[w] H_p = 0$. Polynomial cancellation proves all binary mixed coefficients zero, even at individual rows where $f_h(L) = 0$.

Define finite polynomials

$$\Psi(L) = \sum_{r=0}^m \frac{H_{2r+1}(L)}{(2r+1)!}, \quad g(L) = \sum_{r=0}^m \frac{q_r(L)}{(2r+1)!}.$$

On any binary palette a, b , $\Psi(L)$ thus equals

$$g(L)(f_a(L)a^\Omega + f_b(L)b^\Omega).$$

For two tensors on Ω , let K be the product of local alternating pairings, normalized by $K(a^\Omega, b^\Omega) = 1$. Wick moments with shifted variables $X_i + L_i$ show that $K(\Psi(L), \Psi(M))$ is invariant under simultaneous $\text{SO}(2)$ rotation of the two mean rows: the replica covariance and each determinant are unchanged. Choose independent f_a, f_b . This pairing is

$$(f_a(L)f_b(M) - f_b(L)f_a(M))g(L)g(M).$$

The determinant factor is a nonzero polynomial and is rotation invariant. Cancel it in $\mathbb{C}[D \times D]$, then rotate by 45 degrees and set $M = 0$. Since g is even and $g(0) = 1$,

$$g(L/\sqrt{2})^2 = g(L).$$

If g had positive degree, the degrees of the two sides would differ. Thus $g = 1$, and all q_r with $r \geq 1$ vanish. Together with (3), this proves whole binary vanishing through the terminal degree $2m + 1$. Polarization gives the same statement for products of arbitrary rows from this same D . $\square$

For a source (1), delete any root p . Its three actual incident rows have responses $\tau_h h^{V \setminus \{p\}}$, so their span satisfies (2) with three independent f_h . Thus Lemma 3.1 applies at every original root, before diagonalization.

4 Even omission identities and diagonal reduction

The even-omission argument originates in [8]. Its role is to convert the rooted identities into off-diagonal matrix identities.

Lemma 4.1 (Even omission). *In the setting of (2), omit a vertex q , put $S = \Omega \setminus \{q\}$, and write*

$$\begin{aligned} R &= Q + \sum_h h_q V_h, & L &= U + \sum_h d_h h_q, \\ E(U) &= \sum_{r=0}^m \frac{U^{2r}}{(2r)!} Q^{[m-r]}, & F &= E(0). \end{aligned}$$

For any $L_1, L_2 \in D$ and any active colour k ,

$$[k^S] U(L_1) U(L_2) Q^{[m-1]} = 0.$$

This includes $m = 1$.

Proof. The full coefficient of $q = h$ in $\Psi(L)$ is

$$d_h E(U) + E_{V_h}(U),$$

where subscripts mean directional derivatives in the mean, with Q fixed. By Lemma 3.1, its projection onto any palette h, k on S is $f_h(L) h^S$. All the following pairings use this binary projection implicitly.

Because $|S|$ is even, K is symmetric and $K(Z, h^S) = [k^S] Z$. The same two-replica rotation gives

$$\begin{aligned} K(E(U), E(U)) &= K(F, E(\sqrt{2}U)), \\ K(E(U), E_V(U)) &= \frac{1}{\sqrt{2}} K(F, E_V(\sqrt{2}U)). \end{aligned} \tag{4}$$

The second identity follows by differentiating the rotation identity at means $U, U + tV$: the second rotated mean is zero at $t = 0$ and $E'(0) = 0$. Substitute

$$E_{V_h}(U) = f_h h^S - d_h E(U)$$

and the corresponding identity for $\sqrt{2}L$ into the second equation of (4). The first cancels the d_h terms, leaving

$$f_h(L) [k^S] (E(U) - F) = 0.$$

Cancel the nonzero polynomial f_h . Separating homogeneous degrees and polarizing yields

$$[k^S] U(L_1) U(L_2) Q^{[m-1]} = 0. \tag{5}$$

$\square$

We next apply this identity to the actual hafnian cofactors. This is the diagonal reduction proved in [9].

Lemma 4.2 (Global diagonal reduction). *Every endpoint-colour edge block of a source (1) is diagonal in the given target coordinates.*

Proof. For a fixed colour h , define

$$\begin{aligned} M_h[p, r] &= A_{pr}(h, h) & (p \neq r), \quad M_h[p, p] &= 0, \\ C_h[r, q] &= \text{haf}(M_h[V \setminus \{r, q\}]) & (r \neq q), \quad C_h[q, q] &= 0, \\ B_{ih}[p, r] &= A_{pr}(i, h) & (p \neq r), \quad B_{ih}[p, p] &= 0. \end{aligned}$$

Expansion of the original one-defect word at p gives

$$(B_{ih}C_h)[p, p] = \delta_{i,h}\tau_h.$$

For $p \neq q$, expand each $C_h[r, q]$ at its retained vertex p :

$$\begin{aligned} (B_{ih}C_h)[p, q] &= \sum_{\substack{r, s \in V \setminus \{p, q\} \\ r \neq s}} A_{pr}(i, h) A_{ps}(h, h) \text{haf}(M_h[V \setminus \{p, q, r, s\}]) \\ &= 0. \end{aligned} \tag{6}$$

The last expression is exactly (5) for two rows of the original p family after omitting q . Both sums are ordered; there is no factor $1/2$. Consequently

$$B_{ih}C_h = \delta_{i,h}\tau_h I.$$

Taking $i = h$ proves C_h invertible. Taking $i \neq h$ then proves $B_{ih} = 0$. Every original edge block is therefore diagonal. This uses neither cofactor nonvanishing assumptions nor a change of basis. $\square$

5 The supported-edge identity

The construction in this section is the finite two-replica endpoint argument of [10]. Its inputs are the original root identities obtained above; the auxiliary parameters do not change the edge weights.

Fix two colours B, H after diagonal reduction and roots $p \neq q$. Put $U = V \setminus \{p, q\}$, $N = |U|$ (even), and let Q be the literal binary quadratic on U . Write

$$\begin{aligned} d &= B_{pq}, & e &= H_{pq}, & \beta &= \text{haf}(B), \\ \eta &= \text{haf}(H), & \alpha &= \text{haf}(B[U]), \\ x &= \sum_{i \in U} B_{pi}b_i, & y &= \sum_{i \in U} B_{qi}b_i, \\ u &= \sum_{i \in U} H_{pi}h_i, & v &= \sum_{i \in U} H_{qi}h_i, \\ E(L) &= \sum_{j=0}^{N/2} \frac{L^{2j}}{(2j)!} Q^{[N/2-j]}, & F &= E(0). \end{aligned}$$

Theorem 5.1 (Supported endpoint identity). *For every supported edge $B_{pq} \neq 0$,*

$$\beta = B_{pq} \text{haf}(B[V \setminus \{p, q\}]).$$

Derivatives of E always keep Q fixed. Lemma 3.1, applied to the original p and q row planes and expanded at the other root, gives the exact polynomial identities

$$\begin{aligned} E_y(sx + tu) + dsE(sx + tu) &= s\beta b^U, \\ E_v(sx + tu) + etE(sx + tu) &= t\eta h^U, \\ E_x(ay + bv) + daE(ay + bv) &= a\beta b^U, \\ E_u(ay + bv) + ebE(ay + bv) &= b\eta h^U. \end{aligned} \tag{7}$$

These include the highest coefficient, which uses the terminal original odd response of degree $N + 1$. They are identities in mean parameters on the unchanged source. Differentiating those parameters is legitimate and gives

$$\begin{aligned} E_u(ay) &= 0, & E_v(sx) &= 0, \\ E_{yu}(sx) &= -dsE_u(sx), \\ E_{uv}(sx) + eE(sx) &= \eta h^U, \\ E_{xy}(0) + dF &= \beta b^U. \end{aligned} \tag{8}$$

On tensors on U , use the symmetric binary alternating pairing K , normalized by $K(b^U, h^U) = 1$. The polynomial $K(E(L_1), E(L_2))$ is invariant under the full $O(2)$ action on the two means. The Wick proof is as in Section 3; a determinant -1 transformation contributes $(-1)^N = 1$.

Introduce independent columns $P, Q_c, R_c, S_c \in \mathbb{C}^2$ and means $L_i = P_i x + (Q_c)_i y + (R_c)_i u + (S_c)_i v$. Set

$$\begin{aligned} G &= K(E(L_1), E(L_2)), & f &= G|_{R_c=S_c=0}, \\ M_{ij} &= \partial_{(R_c)_i} \partial_{(S_c)_j} G|_{R_c=S_c=0}, & T &= M + efI. \end{aligned}$$

These are finite polynomials. Orthogonal invariance gives

$$M(OP, OQ_c) = OM(P, Q_c)O^T,$$

and similarly for T . By (8),

$$M_{12} = K(E_u(P_1 x + (Q_c)_1 y), E_v(P_2 x + (Q_c)_2 y))$$

is divisible by $P_1(Q_c)_2$; likewise M_{21} is divisible by $P_2(Q_c)_1$.

Lemma 5.2 (Polynomial orthogonal kernel normal form). *For the kernel matrix just defined,*

$$\begin{aligned} T &= aI + bPQ_c^T, \\ a, b &\in \mathbb{C}[\sigma, \tau, c], \\ \sigma &= P \cdot P, & \tau &= Q_c \cdot Q_c, & c &= P \cdot Q_c. \end{aligned} \tag{9}$$

Proof. In an orthonormal frame where $P_1 = 0$, $M_{12} = 0$, so MP is parallel to P . In a frame where $(Q_c)_2 = 0$, $M_{12} = 0$, so $M^T Q_c$ is parallel to Q_c . Such frames exist on a dense open set of nonisotropic columns. Their eigenvalues agree when $P \cdot Q_c \neq 0$. For two independent columns in dimension two, these conditions force

$$M = a_0 I + bPQ_c^T.$$

Indeed, M minus the common eigenvalue times I annihilates P and has left kernel Q_c ; its rank-one form is a multiple of $(P \cdot Q_c)I - PQ_c^T$. Literal divisibility proves that

$$b = \frac{M_{12}}{P_1(Q_c)_2}, \quad a_0 = M_{11} - bP_1(Q_c)_1$$

are polynomials. The generic identity therefore extends everywhere. Uniqueness and covariance make a_0, b scalar $O(2)$ invariants, and the same is true of $a = a_0 + ef$.

The scalar invariant-ring assertion is the two-vector case of the first fundamental theorem for the orthogonal group; see [4, Section 10.2(a)]. For completeness, we prove this particular case directly. In coordinates $p_+ = P_1 + iP_2$, $p_- = P_1 - iP_2$ and q_+, q_- , $SO(2)$ acts with weights $+1, -1$. Its invariant monomials are generated by $\sigma, \tau, r = p_+q_-, s = p_-q_+$, with $rs = \sigma\tau$. Reflection exchanges r, s . Symmetric polynomials reduce to $r + s = 2c$ and $rs = \sigma\tau$, giving $\mathbb{C}[\sigma, \tau, c]$. These three Gram coordinates are algebraically independent. $\square$

Lemma 5.3 (Polynomial differential equations). *Let R be a commutative integral domain, let $d \in R$ be nonzero, and let $p \in R[z]$. If $p' + dp = 0$, then $p = 0$. If $p' + dp = k$ for $k \in R$, then p is constant and $dp(0) = k$.*

Proof. If $p \neq 0$ has degree r , the coefficient of z^r in $p' + dp$ is d times the leading coefficient of p , which is nonzero. This excludes a homogeneous equation. In the inhomogeneous equation it excludes $r > 0$; substituting a constant polynomial gives $dp(0) = k$. $\square$

The coefficient domain may itself be a polynomial ring. This elementary step, including its endpoint cancellation consequence, is formalized in *PolynomialODE.lean* [12].

Proof of Theorem 5.1. At $(Q_c)_1 = 0$, differentiate M_{12} in $(Q_c)_1$ and use (8). One obtains

$$\partial_{(Q_c)_1} M_{12} \Big|_{(Q_c)_1=0} = -dP_1 M_{12} \Big|_{(Q_c)_1=0}, \quad P_1^2(Q_c)_2(b_c + db) = 0.$$

The substitution

$$(\sigma, \tau, c) = (P_1^2 + P_2^2, (Q_c)_2^2, P_2(Q_c)_2)$$

is injective on polynomial rings: every $\tau \neq 0$ and arbitrary σ, c is attained over $\mathbb{C}$. Thus $b_c + db = 0$. Apply Lemma 5.3 over $\mathbb{C}[\sigma, \tau]$, with polynomial variable c . Since $d \neq 0$, it gives $b = 0$, and hence $T = aI$.

Write

$$Z = E_{uv}(P_2x + (Q_c)_2y) + eE(P_2x + (Q_c)_2y).$$

The full expression is $T_{22} = K(E(P_1x + (Q_c)_1y), Z)$. Differentiate this expression in $(Q_c)_1$, then set $(Q_c)_1 = 0$. Equation (7) gives

$$\partial_{(Q_c)_1} T_{22} \Big|_{(Q_c)_1=0} = P_1 \beta K(b^U, Z) - dP_1 T_{22} \Big|_{(Q_c)_1=0}.$$

The pure- H coefficient of Z is η : positive B mean insertions cannot contribute to an all- H word, so it is the same as at zero, where (8) gives $E_{uv}(0) + eF = \eta h^U$. Thus $K(b^U, Z) = \eta$. Using (9) and the same injective substitution proves

$$a_c + da = \beta\eta.$$

Lemma 5.3, again over $\mathbb{C}[\sigma, \tau]$, gives $a = \beta\eta/d$, independent of all three Gram variables. At the origin,

$$a(0) = K(F, E_{uv}(0) + eF) = \eta K(F, h^U) = \eta\alpha.$$

Since $\eta \neq 0$, this proves

$$\beta = B_{pq} \text{haf}(B[V \setminus \{p, q\}]) \quad \text{whenever } B_{pq} \neq 0. \quad (10)$$

Throughout the argument the edge weights are fixed. All differentiations are in the auxiliary columns P, Q_c , and the polynomial identities are finite. $\square$

6 Every colour is a perfect matching

Corollary 6.1 (Matching rigidity of every active colour). *Each colour graph of a source (1) is a weighted perfect matching.*

Proof. The ordinary hafnian expansion at a vertex p says

$$\beta = \sum_{q \neq p} B_{pq} \text{haf}(B[V \setminus \{p, q\}]).$$

By (10), every nonzero summand equals β . Since $\beta \neq 0$, the number of supported B neighbors of p is exactly one. This holds at every vertex, and for each of the three colours by choosing any other active colour as H . $\square$

Two such matchings cannot share an edge pq : choosing that edge in colour B and the other matching in colour H off p, q gives a mixed word with nonzero coefficient, the product of nonzero matching weights. Consequently their union is a simple cubic graph properly coloured with three colours. Any receiving colour word supports at most one matching, because each vertex has exactly one incident edge of its requested colour. Therefore a mixed perfect matching would have a nonzero coefficient and violate (1).

7 The remaining graph lemma

The following lemma is a special case of the obstruction attributed to Bogdanov in [2, Theorem 7]. At this point, each colour is already a perfect matching, and no vertex colouring can receive cancelling matching contributions. We include a proof for this special case.

Lemma 7.1 (Three-matching obstruction). *Three pairwise disjoint perfect matchings F, G, H on $n > 4$ vertices have a mixed perfect matching in their union.*

Proof. Suppose otherwise. The union $F \cup G$ must be a single alternating Hamilton cycle C : otherwise choosing F on one component and G on the others already gives a mixed matching. Label its vertices $0, \dots, n-1$ cyclically. The matching H consists of chords.

A chord with endpoints of opposite parity leaves two even paths when its endpoints are deleted. Matching those paths along C and adding the chord gives a mixed perfect matching. Hence all H chords join equal parities. The two parity classes must each have even size; otherwise H is impossible.

An even-even chord and an odd-odd chord whose endpoints interlace leave four even paths on deleting their endpoints. Those paths and the two chords give a mixed perfect matching when $n > 4$. Thus no such pair interlaces.

Choose a chord and one of its sides with the fewest vertices strictly inside, over all chords and both sides. Its endpoints have equal parity, so its interior contains a vertex of the opposite parity. Every vertex of that opposite parity in the interior must be H -matched to another interior vertex; a partner outside would give an interlacing opposite-parity chord. Therefore some H chord lies strictly inside the chosen arc. One side of that chord has fewer interior vertices, a contradiction. $\square$

This supplies a mixed perfect matching, contradicting Section 6. Equation (1) is impossible for every even $n > 4$, completing the proof of Theorem 2.1. The graph interpretation in Section 2 and restriction to three active colours now give Theorem 1.2.

8 Boundaries, provenance, and review scope

For $n = 4$, the three matchings of K_4 give the permitted ternary source. In Section 7, a pair of interlacing chords then exhausts the vertices and gives the pure third matching, so no contradiction is asserted. Sections 3–6 remain valid at $n = 4$, including their terminal response coefficients. For $n = 2$, the odd-core lemma does not apply and arbitrary target dimension is possible. Odd n has no perfect matchings in this model.

The historical sources [5, 6, 7, 8, 9, 10] are internal research notes from this repository. Their proofs have been included here, so these citations identify provenance rather than replace mathematical steps. The published external inputs are the graph formulation and combinatorial obstruction [1, 2]; the pairing and invariant-theory references [3, 4] provide background for identities also proved in the text.

The internally reviewed mathematical source is the frozen all-orders proof in [11]. Its two full-chain audits were carried out by separate research agents in this project. They are internal reviews, not independent external peer review. The present manuscript reorganizes that argument, adds graph definitions and bibliographic references, and states the polynomial-degree argument as Lemma 5.3.

The Lean package in [12] checks eight algebraic lemmas: the polynomial differential-equation statements and the finite-sum implication from the supported-edge identities to degree one. The source-to-kernel construction, the whole-response identities, the invariant-theory reduction, and the final graph argument are not yet formalized there. Thus the Lean package does not verify the full conjecture.

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